

Predicting Heart Disease: Insights from Logistic Regression and Random Forest

Data 621 Final Project

Mohamed Hassan-El Serafi, Chun Shing Leung, Eddie Xu, Keith Colella, Yina Qiao

Prof. Jeffrey L. Edwards

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Abstract

This study evaluates the effectiveness of Logistic Regression and Random Forest models in predicting heart disease using a small dataset of 13 medical variables. The models were assessed based on sensitivity, specificity, and interpretability, aiming to balance predictive performance with practical application in healthcare settings. Logistic Regression emerged as the preferred model, offering interpretable coefficients that quantify the impact of key predictors, including vessel blockages, chest pain, exercise-induced angina, and gender. While Random Forest demonstrated slightly higher sensitivity and specificity, its complexity limits its utility in resource-limited or clinical environments where explainability is critical.

The study underscores the importance of actionable insights provided by Logistic Regression, enabling healthcare providers to better assess and mitigate heart disease risk. However, the analysis is constrained by the dataset's small size, which may affect the generalizability of findings. Future research should consider larger, more diverse datasets, integrating additional factors such as genetic predisposition, lifestyle habits, and family history to enhance model robustness. Alternative machine learning approaches, such as Gradient Boosting Machines or Elastic Net Logistic Regression, could also be explored to achieve a balance between accuracy and interpretability. Additionally, exploring the ability to not only predict disease presence but also its severity might further support clinical applications.

Keywords

Heart Disease Prediction
Classification Models
Logistic Regression
Random Forest

Introduction

Heart Disease is one of, if not the most common, causes of death worldwide. The World Health Organization (WHO) estimates that heart disease accounts for 17.9 million global fatalities annually. According to the Center for Diseases Control (CDC), heart disease has been the leading cause of death in the United States since 1921. The National Institutes of Health (NIH) states that cardiovascular diseases (CVDs) accounted for 20.5 million deaths in 2021, comprising approximately one-third of all global deaths. There are many risk factors associated with heart disease. The CDC has highlighted high blood pressure, high cholesterol, and smoking as key risk factors. While other risk factors such as age and family history cannot be controlled, other lifestyle factors such as smoking cigarettes, drinking alcohol and poor diet can be attributed to an increased risk for the disease. Despite advances in technology that have enhanced our ability to detect and diagnose heart disease, there are still uncertainties of what factors to pinpoint that can help our understanding of what signs to look for.

There are different types of cardiovascular diseases. Here are some common examples:

- * **Coronary Artery Disease (CAD):** the most common type of heart disease, CAD occurs when there is plaque buildup in the arteries that block blood supply to the heart. An artery is a major blood vessel that transports blood to the heart. Over time, the walls of the arteries can become clogged with fatty deposits. This process is called atherosclerosis and the fatty deposits are called atheroma. This prevents the arteries of the heart from delivering enough oxygen-rich blood to the heart.
- * **Peripheral Arterial Disease (PAD):** This develops when arteries feeding the limbs become narrow, reducing blood flow from the heart to the arms and legs. This is usually felt in one or both knees, hips, and calves. The major cause is the buildup of fatty plaque in the arteries (atherosclerosis). PAD can damage any blood artery; however, it more commonly affects the legs than the arms.
- * **Myocarditis:** It is an inflammation of the inner muscles of the heart caused by a variety of parasitic and microbial infections. Infections with microorganisms, fungi, parasites, viruses (most often, viruses that cause the flu virus, influenza, or COVID-19), or any other microorganisms can induce myocardial inflammation. Autoimmune diseases can also trigger myocarditis due to the immune system's ability to target any organ in the human body, along with the heart, and cause inflammation.
- * **Congenital heart disease:** It is a condition that is associated with one or even more structural cardiovascular issues which have occurred since birth. It changes the nature of blood through the heart from birth. Congenital heart disease symptoms may not develop until the patient is an adult.
- * **Arrhythmia:** an irregular heartbeat. If a person has this condition, their heart may beat excessively rapidly, extremely slow, very early, or in an irregular rhythm. This occurs when the electrical impulses that control heartbeats fail. An irregular heartbeat can feel like a rushing or fluttering heart.

The objective of this study is to build classification models that can help predict heart disease based on variables in the dataset. We chose this dataset because we believe it incorporates variables that are pertinent to understanding how we can better predict heart disease. There are 1,025 participants in the study, involving 13 different features. The data utilizes information from participants located in Cleveland, Hungary, Switzerland, and Long Beach V. Independent variables in the dataset that will be explored include age, gender, cholesterol, Chest Pain levels (0-3), resting blood pressure, whether or not exercise-induced angina is present, maximum heart rate achieved, oldpeak, number of major vessels colored by fluoroscopy, the slope of the peak exercise ST segment, fasting blood sugar, resting electrocardiographic results, and thallium heart scan results. We will explore the relationships between the independent variables and the dependent variable, as well as the relationships with the independent variables among each other. The types of models used for this study were Logistic Regression and Random Forest. We will explain our reasoning for choosing these methods, as well as the metrics we used to evaluate the performance of each model and ultimately the model we chose to select. Finally, we will explain further what we should take away from the results, factors not in the dataset that should be considered in future studies that could potentially provide further clarity of detecting heart disease, and other areas concerning heart disease that can be built upon our models for future exploration.

Literature Review

There are numerous studies that explore heart disease prediction using machine learning models. In fact, many machine learning models including Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN) have been evaluated and compared with each other to determine the most suitable model for heart disease predictions.

As per Yazdani A, Varathan KD, Chiam YK, Malik AW, Wan Ahmad WA. (2021) paper, the team explores machine learning techniques to apply on heart disease prediction and decides to use a data mining technique called Weighted Association Rule Mining (WARM) to explore relationships between features and generate predictions using data mining rules. The purpose is to compute the weight of significant features in heart disease prediction, which can contribute to a higher accuracy. This is different from our approach since the WARM model is applied and compared with other WARM models. In our approach, we are using classifier models instead of data mining techniques. The authors of the WARM model applied the Apriori algorithm to execute the technique on their data. In the algorithmic process of Apriori, an item set X of length k is frequent if and only if every subset of X , having length $k-1$, is also frequent. According to the authors, this consideration results in a substantial reduction of search space and allows rule discovery in a computationally feasible time. Additionally, the continuous features used in their study were categorized and given different ranges. For instance, age was categorized into three classifications: participants under 40, between 41 and 64, and those 65 or older. Blood pressure was classified as normal, unusual and high, and cholesterol was categorized as normal, borderline high, high, and very high. This also differs from the approach used in this study,

where the continuous variables were not transformed into categorical variables and instead preserved as continuous variables.

A study conducted by El-Sofany, H., Bouallegue, B. & El-Latif, Y.M.A. (2024) utilizes XG Boost to predict heart disease in participants. The authors prepared the data by using StandardScaler to normalize all the features to the relevant coefficient, ensuring each feature has a zero mean and one variance. Next, they applied the Synthetic Minority Oversampling Technique (SMOTE) to overcome the imbalance in the data. For selecting features, the authors employed various techniques, highlighting analysis of variance (ANOVA), chi-square, and mutual information (MI) as feature selection methods employed. The authors used 10 distinct Machine Learning algorithms, including Naive Bayes, SVM, voting, XGBoost, AdaBoost, bagging, Decision Tree (DT), KNN, RF, and LR, evaluating the performance measures of the models using true positive (TP), true negative (TN), false positive (FP), and false negative (FN) data. Among the models used, the authors found XG Boost as the best performing algorithm for predicting heart disease in patients, with an accuracy of 97.57%.

In most studies, it shows that the Logistic Regression model may be the most suitable model for heart disease prediction. As per Anshori, M., & Haris, M. S. (2022) paper, the logistic regression model was applied and compared with other two classifier models, Support Vector Machines (SVM linear) and Linear Discriminant Analysis (LDA), to determine its effectiveness. This is similar to our approach of comparing the Logistic Regression with another classifier model. However, unlike our approach, their number of features is set to 9 while our number of features is 13. Aside from that, their model selection is different from our selection. SVM and LDA were selected and the end result shows both models are outperformed by the Logistic Regression in terms of accuracy and computational times. We selected Random Forest to compare with Logistic Regression instead due to its simplicity, high accuracy, and faster computational times.

In fact, a study conducted by Pal, M., & Parija, S. (2021) reviewed and evaluated the effectiveness of Random Forest Model. They used the same data set with 14 features and applied 10-folds cross validation technique to split data for training and testing purposes. The Random Forest model was processed and as a result, their model achieved an accuracy of 86.9%. Their approach is different from our approach toward Random Forest and model prediction. In our approach, we evaluated the model first and then refined the model further by identifying the important features. Besides that, there was no performance comparison to compare their Random Forest model with other models to assess their performance. We compared our model with a Logistic Regression model to see the difference in performance.

Research Question

Which model—Logistic Regression or Random Forest—offers the most suitable approach for predicting heart disease based on medical test data?

Overview

Exploratory Data Analysis

To understand the dataset and ensure its readiness for modeling, we conducted an extensive exploratory data analysis:

1. **Data Type Inspection:** Each variable was reviewed and categorized as either numerical or categorical to ensure appropriate treatment during modeling.
2. **Missing Value Assessment:** The dataset was checked for missing values, confirming the data's completeness.
3. **Correlation Analysis:** A correlation matrix was used to identify potential strong linear relationships among numerical variables, ensuring multicollinearity did not unduly affect regression results.
4. **Distribution and Balance Checks:** Using bar plots, histograms, and frequency plots, we examined the distribution of variables to detect imbalances in the dataset, ensuring model fairness and effectiveness.

Data Preparation for Classification Modeling

After EDA, we prepared the data with the following steps:

1. Variables were appropriately transformed to match their data types (categorical or numerical).
2. Features were analyzed for multicollinearity using Variance Inflation Factor (VIF) checks.
3. The dataset was split into training and testing subsets, maintaining consistency across models for fair comparison.

Model Building

We developed and compared two models: Logistic Regression and Random Forest Classification Model.

Logistic Regression Model:

1. A full model including all variables was initially built.
2. Multicollinearity was addressed using VIF checks, and features were further refined using Stepwise AIC selection to prioritize prediction accuracy.
3. Coefficients and marginal effects were calculated to assess the influence of each independent variable on the target variable.

McFadden's Pseudo R-squared was used to evaluate the model's fit.

The final logistic regression model was used to predict the target variable in the test dataset, with accuracy assessed via a confusion matrix.

Random Forest Model:

1. A full model was constructed, and error rate plots guided the selection of an optimal number of trees.
2. Feature importance was visualized to identify key predictors and refine the model.
3. The final Random Forest model was tested on the same dataset, with performance evaluated through a confusion matrix.

Model Selection

Both models were evaluated using Receiver Operating Characteristic (ROC) curves and the Area Under the Curve (AUC). The model with the better balance of sensitivity and specificity was selected as the superior model, demonstrating its practical applicability for heart disease prediction.

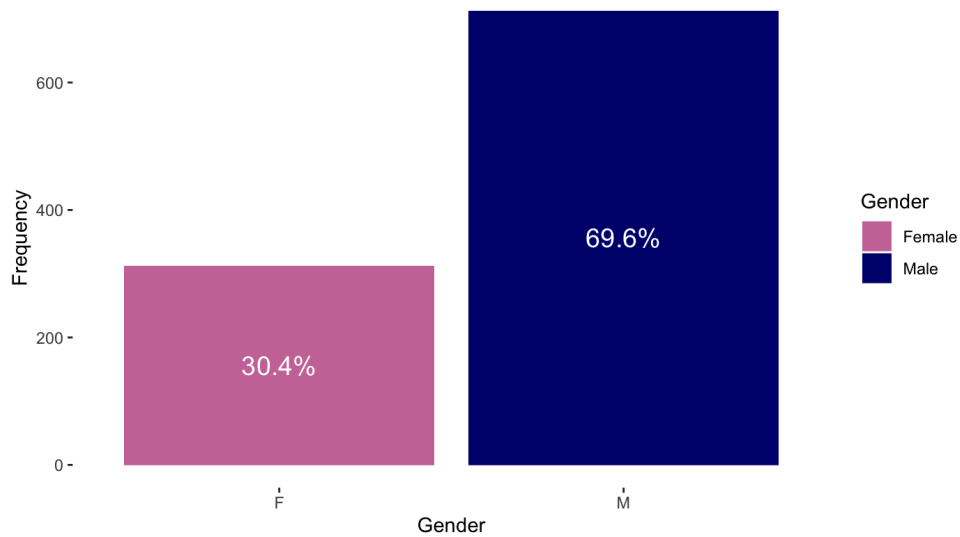
Experimentation and Results

Variable Overview

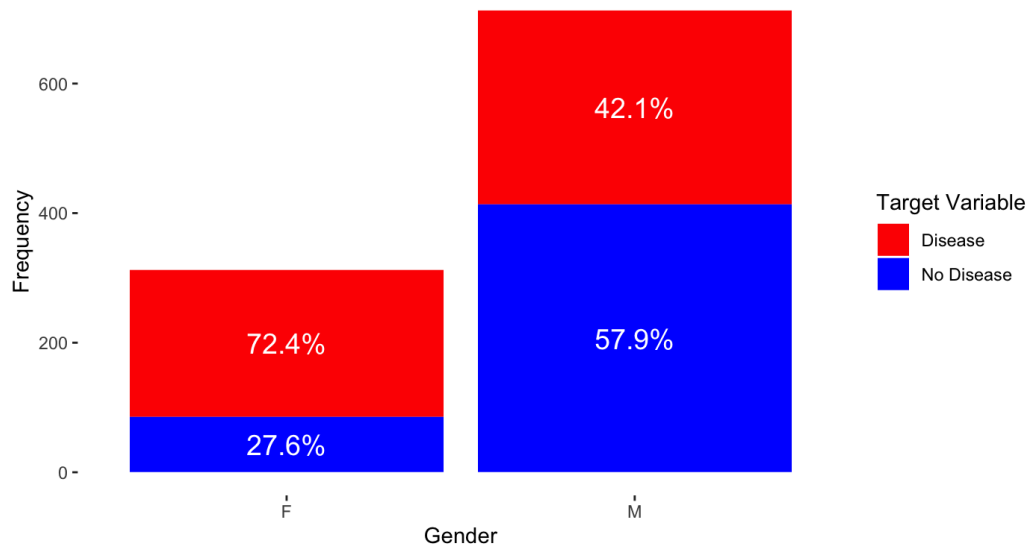
Male and Female Participant Distribution

Signs and symptoms of heart disease are different between men and women. While chest pain is a discernible symptom among men, women have less obvious symptoms. While women may also experience chest pressure, other symptoms they may also report include nausea, sweating, vomiting, and pain in the neck, jaw, throat, abdomen or back. Additionally, women have smaller blood vessels and arteries, and the muscular walls of women's hearts are thinner. Therefore, it can be more difficult to identify heart disease and cause delays in treatment.

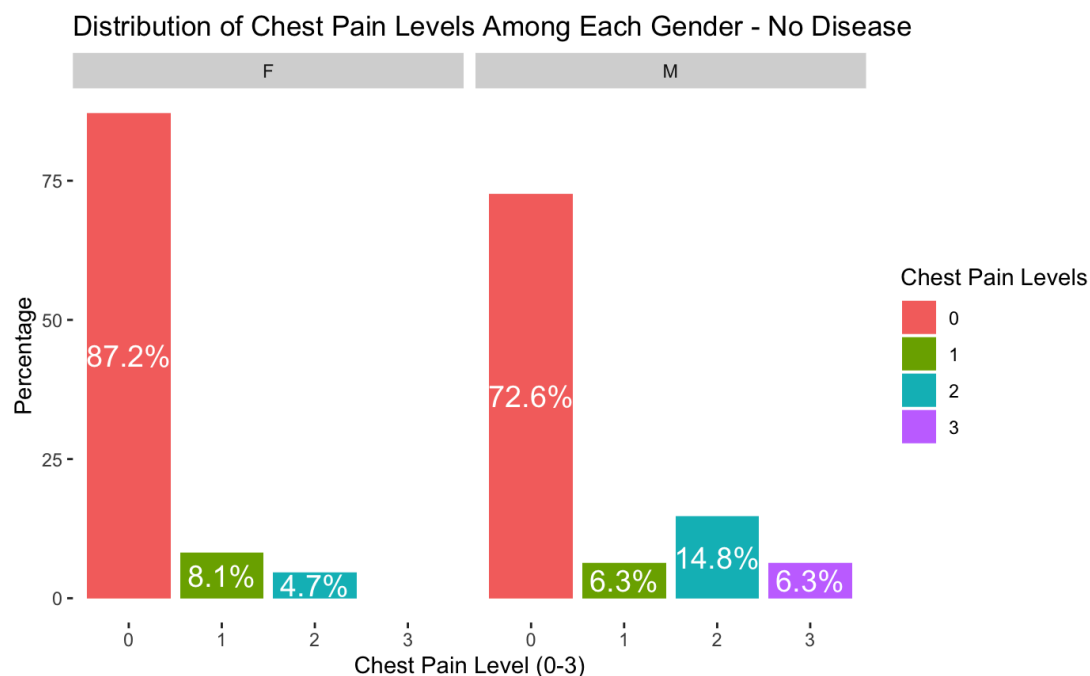
Percentage of Male and Female Study Participants



Frequency and Percentage of Participants With and Without Heart Disease for Males and Females



The participants in the study are almost 70% male, and should be taken into consideration when used as an independent variable to predict heart disease. When analyzing the breakdown of male and female participants with and without heart disease, 72.4% of female participants had the disease, an extraordinarily high percentage given the small number of female participants. Conversely, almost 58% percent of male participants did not have heart disease. This is significant, since diagnosing heart disease in women is more difficult than in men, for reasons previously stated.

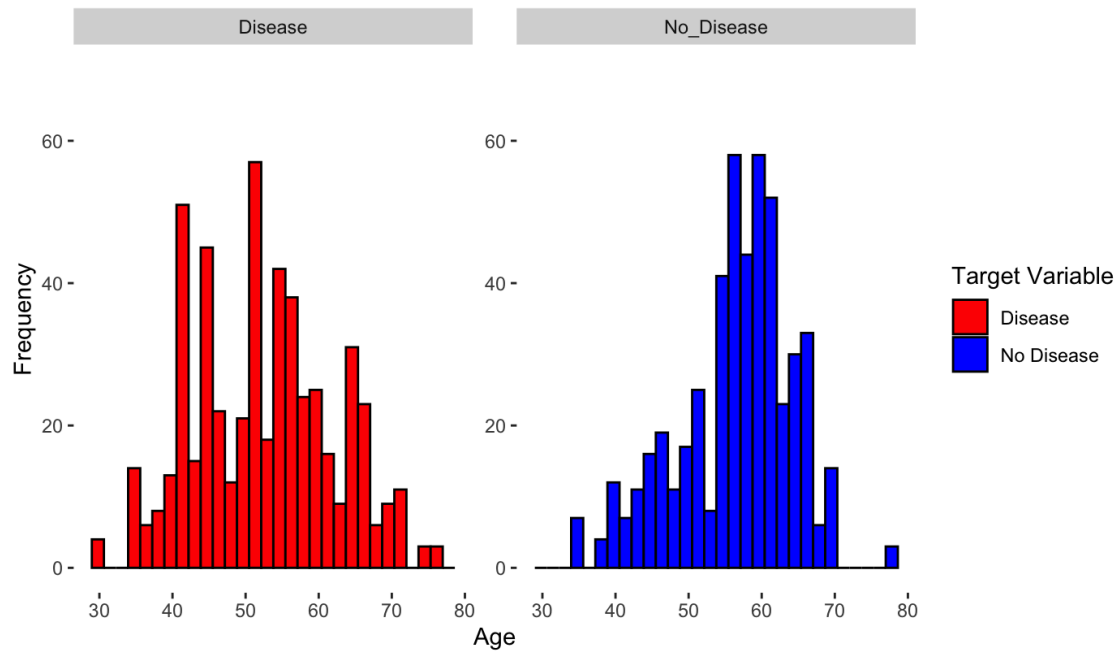


The breakdown of males and females with and without heart disease by different chest pain levels brings into further focus how symptoms vary by gender. Unsurprisingly, for participants without heart disease, 72.6% of males and 87.2% of females reported a level 0 chest pain. When analyzing participants with heart disease, the breakdown of chest pain levels between males and females reveals interesting observations. 46.5% of female participants reported a chest pain level of 2, the highest percentage among all levels reported among all participants with heart disease. 38% of males with heart disease reported a chest pain level of 2 as well, which was the 2nd highest percentage among all levels reported among all participants with heart disease. When comparing the highest level of chest pain (3) between males and females, 12.7% of males reported a chest pain level of 3, compared to 5.8% of females who reported the same chest pain levels. This is important to highlight, illustrating how males can exhibit more pronounced chest pain levels compared to their female counterparts.

Age Distribution

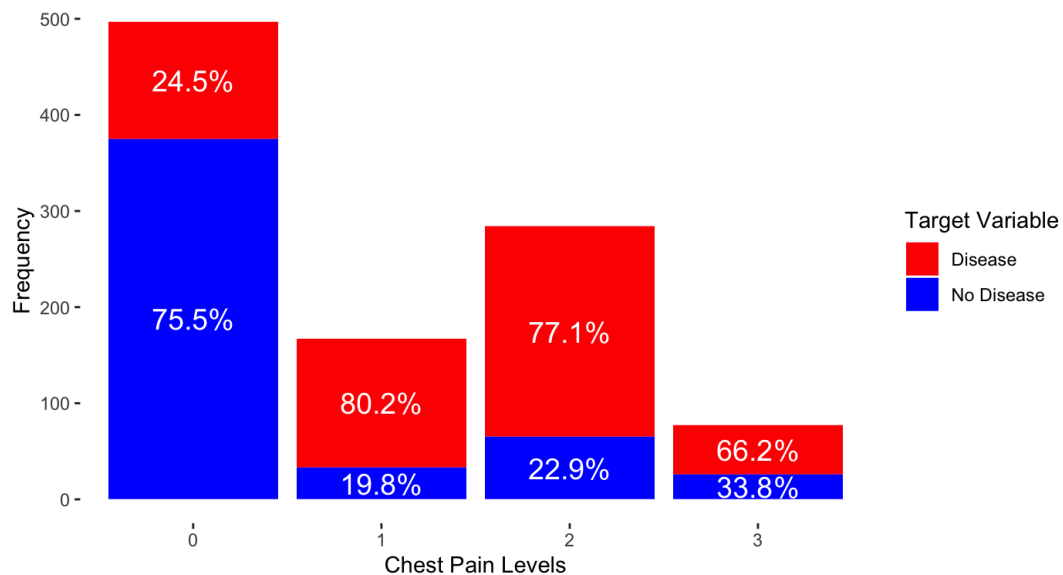
For participants with heart disease, there appears to be higher frequencies of participants who have the disease around their 40s and 50s. Those without the disease had higher frequencies of participants with ages in their late 50s to 60s. The age of participants with the disease shows slightly more spread than those without the disease.

Age Distribution of Participants With and Without Heart Disease



Chest Pain Levels

Frequency and Percentage of Participants With and Without Heart Disease at Each Chest Pain Level

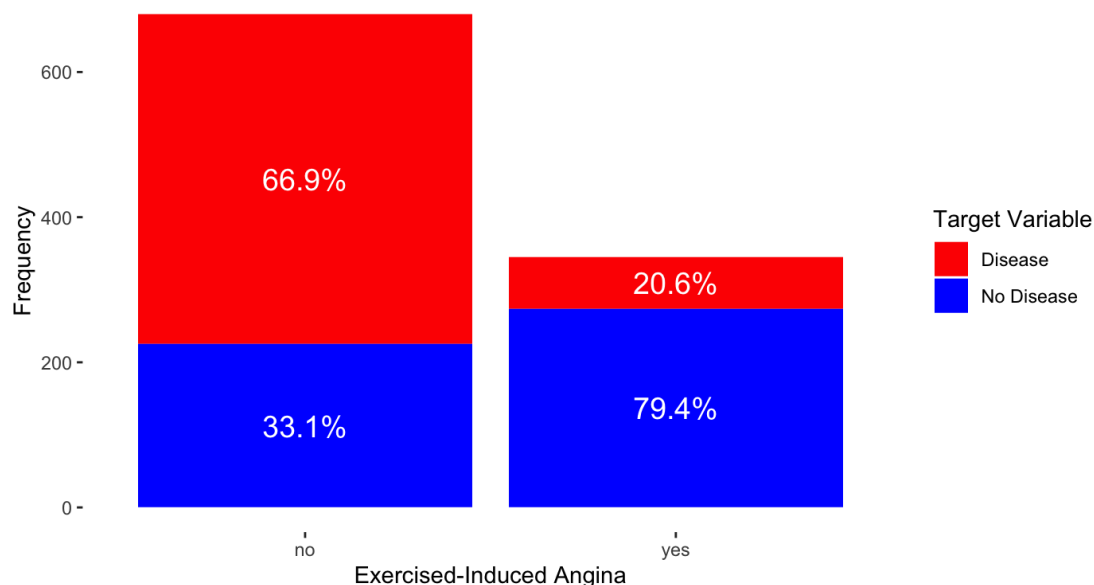


Participants reporting chest pain levels 1, 2 and 3, respectively, overwhelmingly were diagnosed with heart disease. The percentage of participants with the disease that reported chest pain levels 1, 2, or 3 declined as the level increased, from 80.2% of heart disease participants reporting level 1 chest pain to 77.1% with level 2 chest pain and 66.2% with level 3. Among participants reporting 0 level chest pain, almost a quarter had the disease, which may indicate that variables

other than chest pain may be better indicators for heart disease who aren't exhibiting chest pain.

Exercise-Induced Angina

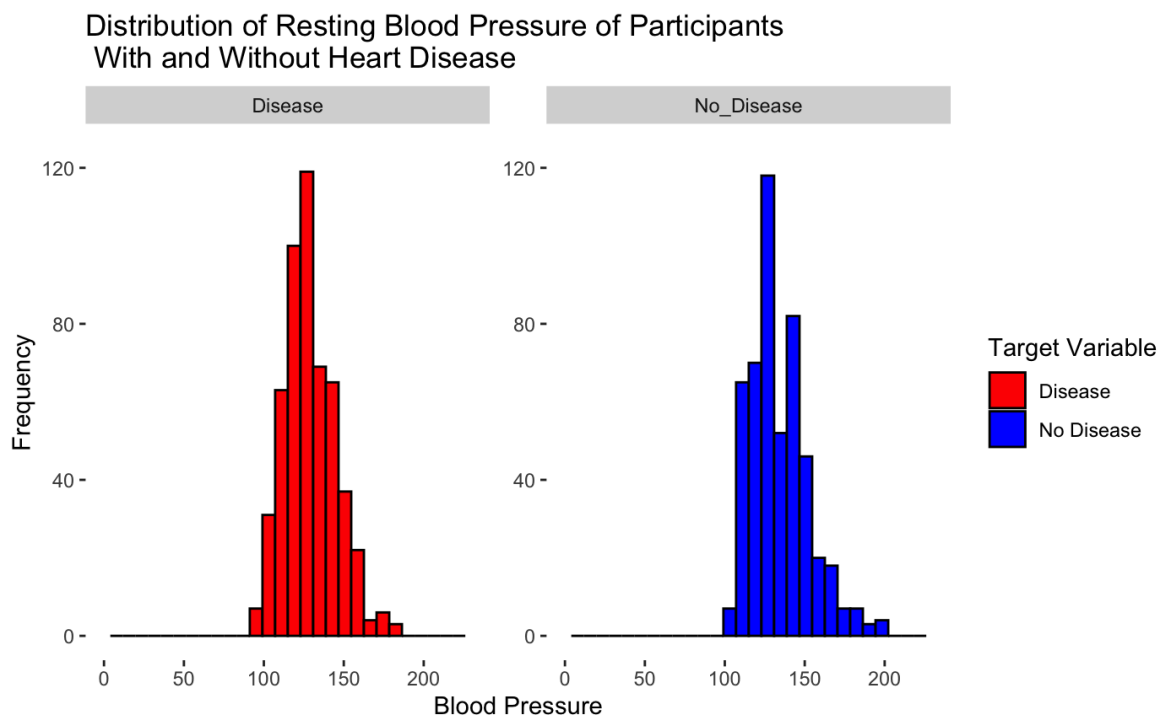
Frequency and Percentage of Participants With and Without Heart Disease That Have and Don't Have Exercised Induced Angina



Participants who experienced angina while exercising and were diagnosed with heart disease made up around 20%. Alternatively, around 67% of participants who did not experience angina while exercising were diagnosed with heart disease. These statistics may indicate that angina may not be a factor when diagnosing heart disease.

Resting Blood Pressure, Cholesterol and Age

Blood pressure and cholesterol are perceived to be two indicators of heart health. Blood pressure measures the force in the arteries as the heart pumps, while cholesterol is a waxy, fat-like substance found in the blood and can impact heart health if there are high levels discovered. Because each variable is a measurement of the performance of blood vessels in connection with the heart, we decided to look at each variable separately and their relationship with one another and with age. Do participants with and without heart disease have different blood pressure and cholesterol levels, and do these levels vary with age?

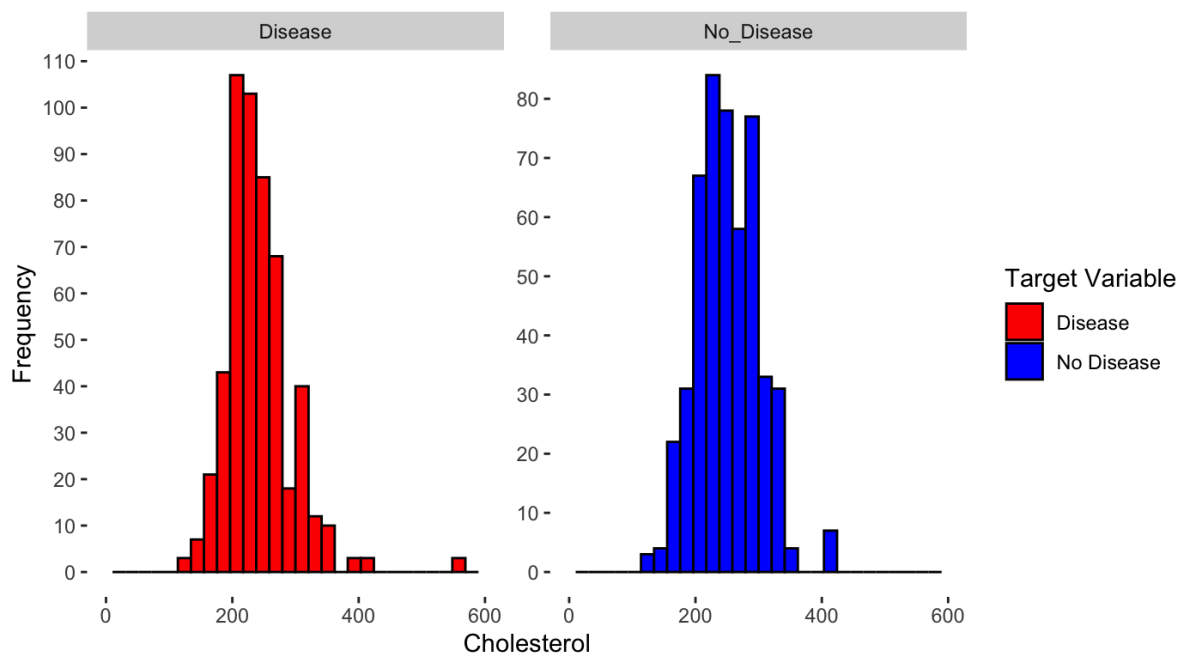


When analyzing the distribution of participants with and without heart disease by resting blood pressure, there weren't discernible differences, with a couple of slight exceptions. The number of participants without heart disease showed skewness to the right, and the range of blood pressure was slightly wider for that group. The center distribution for those with the disease appears to be around a lower blood pressure value than the center of the distribution for those without the disease. This also appears to be the case when comparing the frequency of cholesterol values for those with and without heart disease, with the center of the distribution for those with the disease aligning with a value of around 200, slightly lower than that depicted for those without the disease.

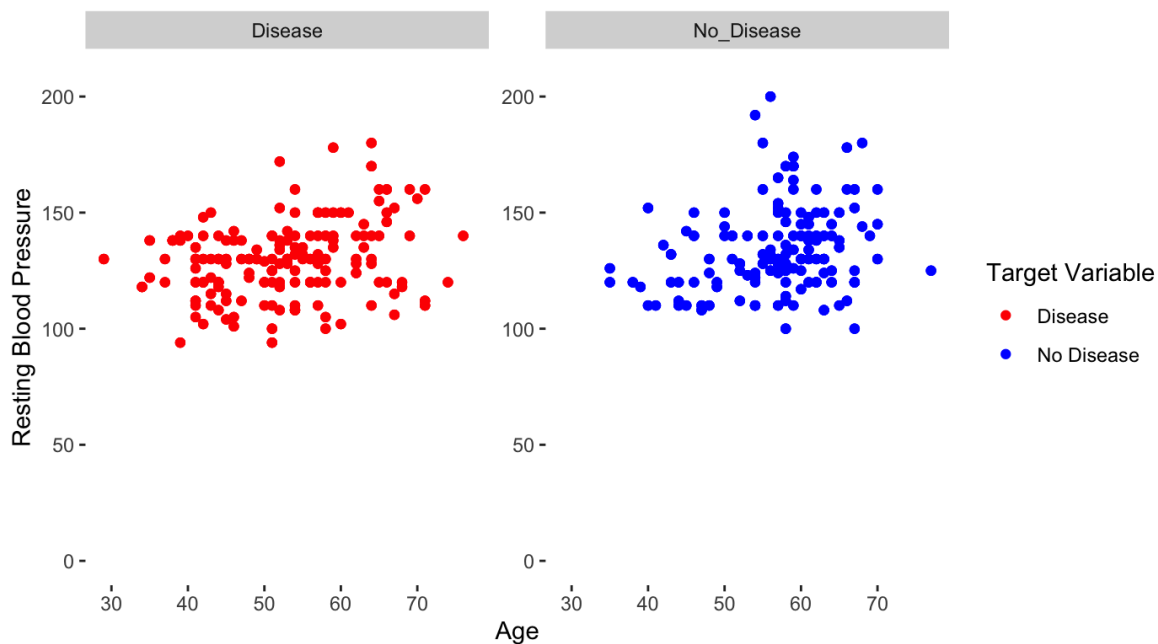
We compared resting blood pressure with age and cholesterol, respectively, for those with and without heart disease. There was a cluster of participants with the disease between the ages of 40 and 60 and blood pressure levels between 100 and 150. Conversely, for those without the disease, there was slightly less of a cluster and more spread of age and blood pressure levels, including outliers of people with blood pressure levels at or near 200 and non-disease participants in their 70s.

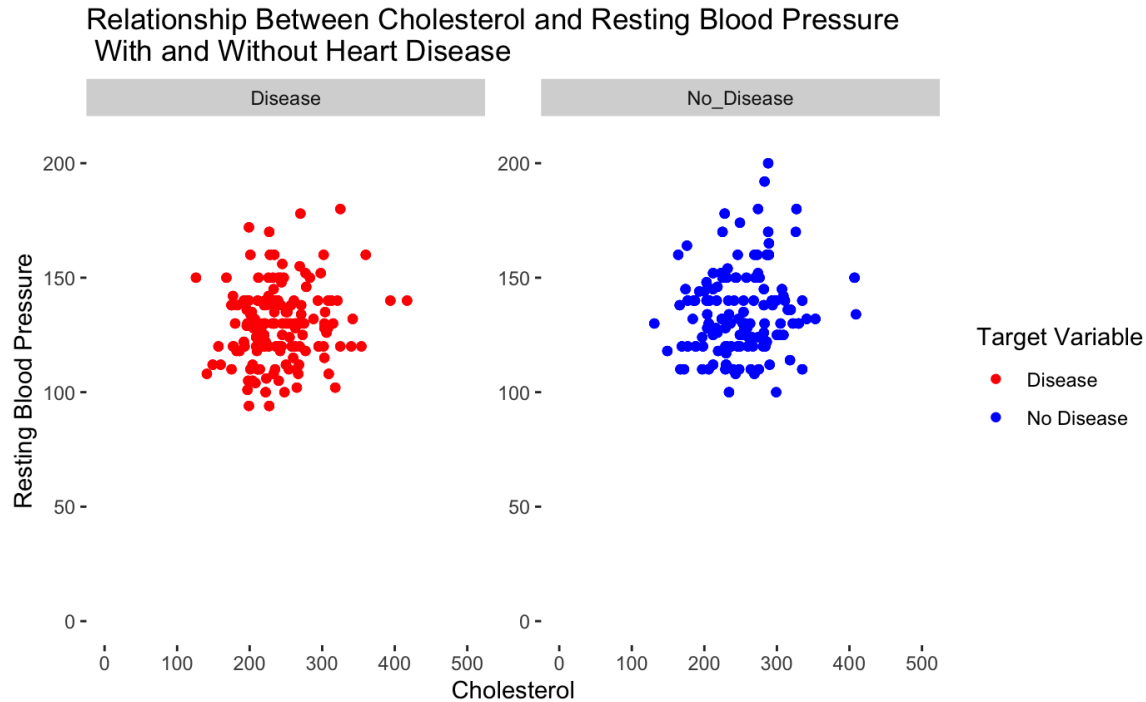
The relationship between blood pressure and cholesterol followed a similar pattern for blood pressure and age. Participants with the disease were clustered between blood pressure values 100 and 150, and between 200 and 300 cholesterol values. Those without the disease showed slightly more spread with the same blood pressure outliers as previously noted and outliers of participants with cholesterol levels around 400.

Distribution of Cholesterol of Participants
With and Without Heart Disease



Relationship Between Age and Resting Blood Pressure
With and Without Heart Disease

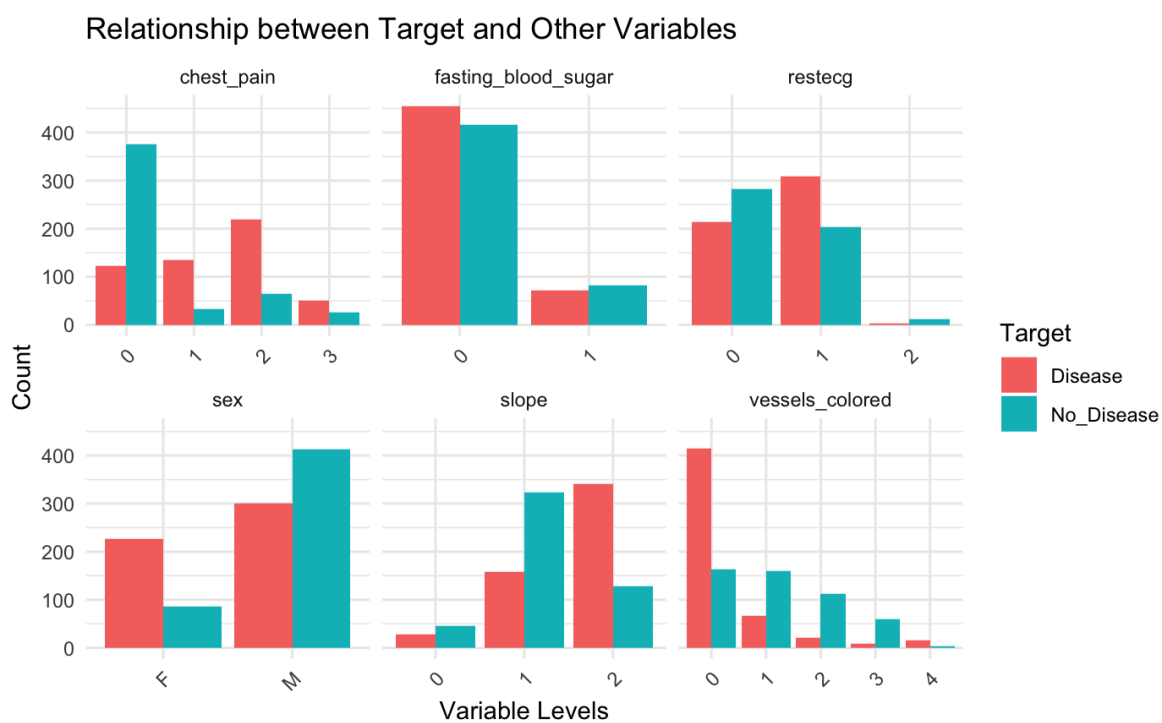
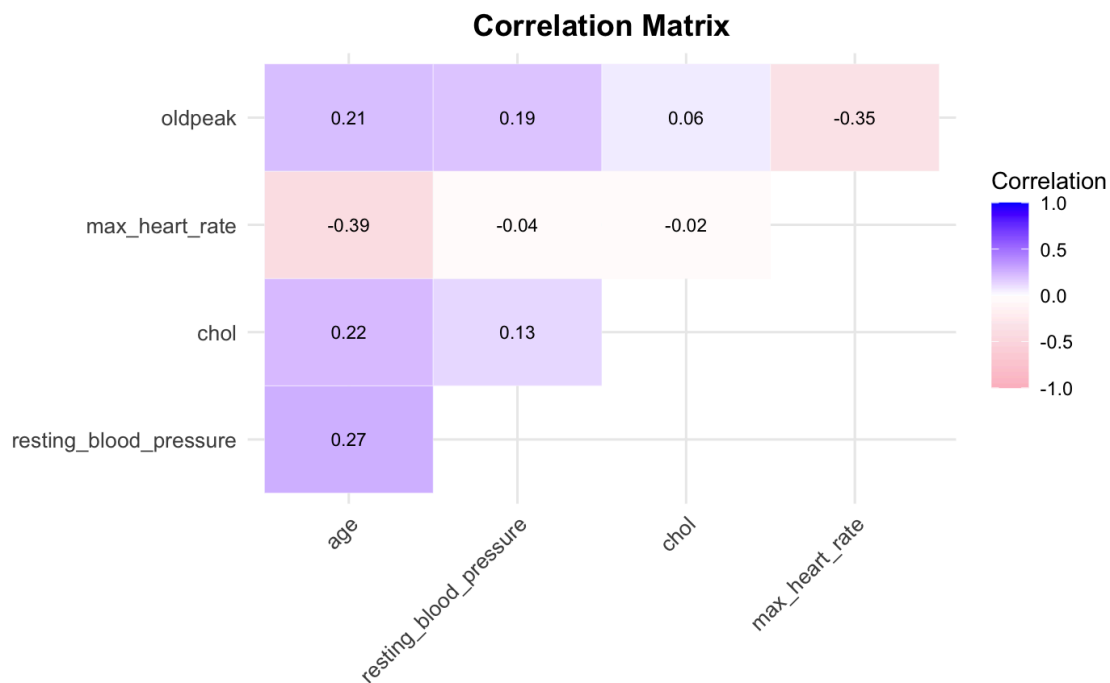




Because there doesn't appear to be a discernible difference for those with and without the disease, blood pressure, age and cholesterol, taken separately, may not have a strong correlation with the target variable. Additionally, blood pressure doesn't appear to be strongly correlated with age and cholesterol, which is important when considering multicollinearity among the independent variables.

Exploratory Data Analysis

The exploratory data analysis revealed no missing values in the dataset and a balanced distribution across classes, ensuring no significant imbalance issues. The correlation matrix indicated that the variables exhibited weak or no linear relationships (correlation coefficients ranging between 0.1 and 0.3 or close to zero). Variance Inflation Factor (VIF) scores ranged between 1 and 5, showing moderate correlation among the variables, which is not a major concern. While multicollinearity in logistic regression can inflate standard errors and make coefficient estimates unreliable, the VIF scores suggested no significant issues, allowing us to proceed with model building.



	GVIF	Df	GVIF ^{1/(2*Df)}
age	1.505194	1	1.226863
sex	1.672999	1	1.293445
chest_pain	1.944615	3	1.117221
resting_blood_pressure	1.250533	1	1.118272
chol	1.330823	1	1.153613
fasting_blood_sugar	1.158931	1	1.076537
restecg	1.182297	2	1.042754
max_heart_rate	1.578211	1	1.256269
exercise_induced_angina	1.197461	1	1.094286
oldpeak	1.540111	1	1.241012
slope	1.848202	2	1.165970
vessels_colored	1.987327	4	1.089642
thal	1.516112	3	1.071820

Logistic Regression Model

We prioritized prediction accuracy by using the Akaike Information Criterion (AIC) for feature selection. The StepAIC function reduced the model complexity, producing a final model with the lowest AIC score (516.19). Two variables, fasting blood sugar and resting blood pressure, were removed during this process.

One of the strengths of logistic regression is its interpretability. We calculated the coefficients and marginal effects to understand the impact of independent variables on the likelihood of heart disease. However, coefficients in logistic regression are scaled by a factor, so their magnitude does not directly indicate the size of their effect. Instead, they help identify which variables are more likely to influence heart disease risk. For example:

Male individuals, higher cholesterol levels, exercise-induced angina, higher old peak values, and specific levels of chest pain type or major vessel coloring significantly increased the likelihood of heart disease.

Marginal Effects provide a more intuitive interpretation by quantifying the change in the probability of the target outcome for a one-unit change in the predictor, holding all other variables constant. Examples include:

Males: 20.76% more likely to have heart disease.

Vessels colored 1, 2, 3: 18.46%, 30.64% and 18.17% more likely, respectively.

Chest pain type 1: 9.86% less likely.

Chest pain types 2 and 3: 17.37% and 22.06% less likely, respectively.

Vessels colored 4: 15.77% less likely.

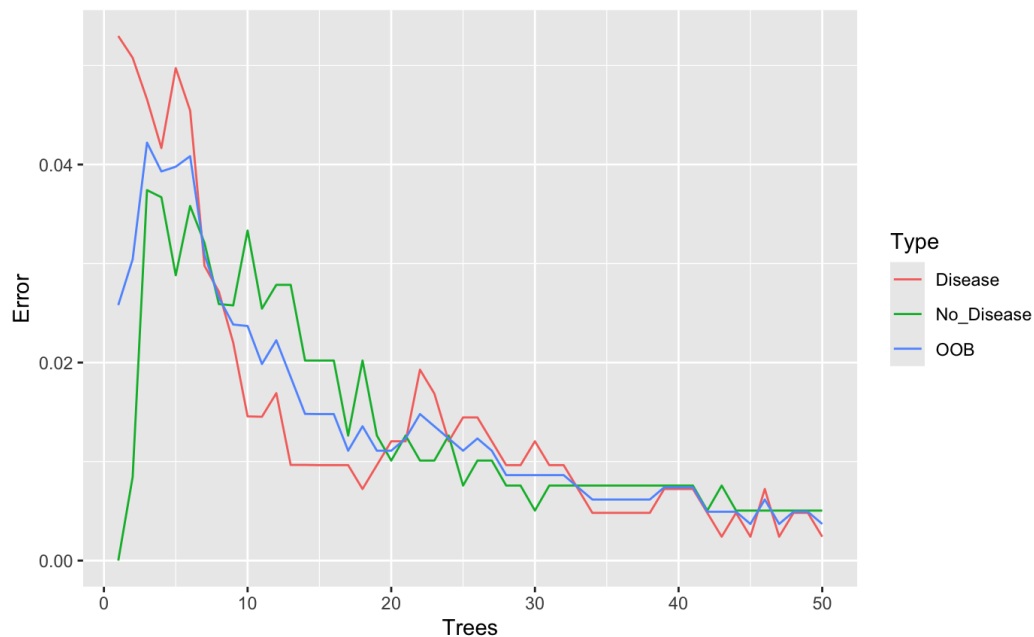
(Intercept)	sexM	chest_pain1	chest_pain2
-0.0538679592	0.2075653217	-0.0985720709	-0.1737605650
chest_pain3	resting_blood_pressure	chol	max_heart_rate
-0.2206124916	0.0016709235	0.0006260614	-0.0021264326
exercise_induced_anginayes	oldpeak	slope1	slope2
0.0440599095	0.0332853043	0.0542928138	-0.0642742110
vessels_colored1	vessels_colored2	vessels_colored3	vessels_colored4
0.1845652488	0.3063930696	0.1817143268	-0.1575724922
thal1	thal2	thal3	
-0.3016922167	-0.2457665238	-0.1168083548	

We also assessed model fit using McFadden's Pseudo R-squared, which measures how well the model explains the data relative to a null model. Our logistic regression model achieved a McFadden's R-squared of 0.5745, a strong result indicating that the model explains approximately 57.45% of the variation in the outcome variable. Typically, values between 0.2 and 0.4 are considered acceptable, making this an exceptionally good fit.

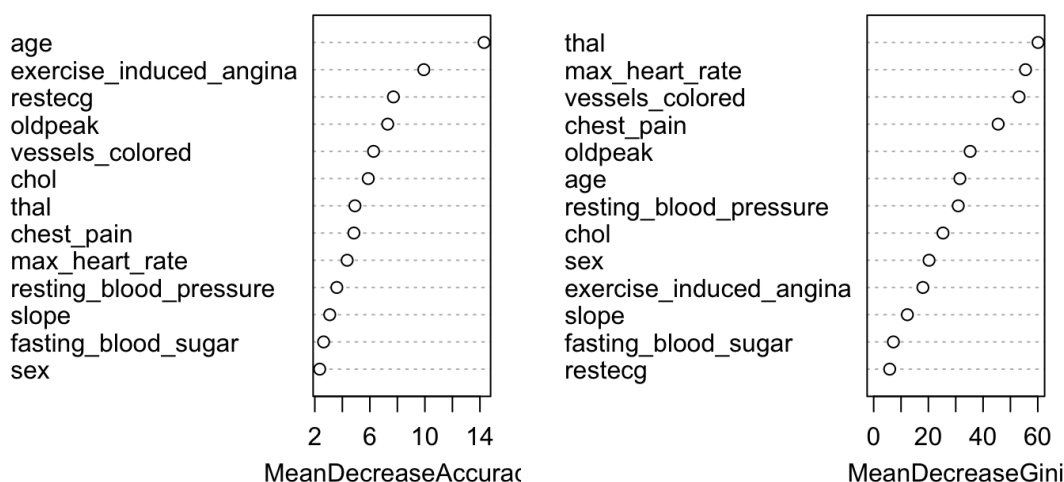
Random Forest Classification Model

The Random Forest model was implemented as a classification model, given the categorical nature of the target variable. Initially, the model exhibited a very low out-of-bag (OOB) error rate, signaling potential overfitting. To address this, we:

1. Reduced the number of trees, as the error rate plot suggested diminishing returns after 5 trees.
2. Performed feature selection by analyzing variable importance. Variables such as fasting blood sugar, resting blood pressure, and slope were removed due to their low importance, improving model efficiency. The sex variable was retained due to its critical role in understanding gender-specific risks.



rf2



Model Comparison and Evaluation

Logistic Regression is a binary classification model that predicts probabilities and is evaluated using McFadden's Pseudo R-squared, based on likelihood ratios. Random Forest, a non-linear ensemble method, focuses on reducing classification errors and out-of-bag (OOB) error, with no direct equivalent to R-squared. Thus, McFadden's Pseudo R-squared in Logistic Regression cannot be directly compared to Random Forest's performance metrics.

For classification tasks like this, performance metrics such as ROC-AUC and confusion matrix-derived measures are more relevant for comparing models like Logistic Regression and Random Forest. These metrics assess how well the models predict outcomes, rather than how much variance is explained.

To evaluate the predictive performance of the Logistic Regression and Random Forest models, we:

1. **Confusion Matrices:** We began by generating confusion matrices to assess classification accuracy and provide deeper insights into each model's behavior, particularly focusing on sensitivity and specificity.
2. **Sensitivity and Specificity:**
 - Sensitivity (True Positive Rate) measures the ability of the model to correctly identify disease cases. Here, Random Forest excels with a perfect sensitivity of 1.0000, meaning it correctly identifies all actual disease cases, which is crucial when failing to detect a disease (false negatives) is critical.
 - Specificity (True Negative Rate) assesses the ability to identify non-disease cases. Random Forest also outperforms Logistic Regression in this regard with a

specificity of 0.9417 compared to 0.7961, which is important for minimizing false positives.

3. Kappa Statistic:

- The Kappa statistic measures agreement between observed and predicted classifications beyond chance. Logistic Regression shows a Kappa of 0.7088, indicating substantial agreement, while Random Forest has a slightly higher Kappa of 0.9437, which suggests very strong agreement. However, both models demonstrate reliable classification performance.

4. ROC Curve and AUC:

- Random Forest has a higher AUC of 98.5%, indicating a superior ability to distinguish between disease and non-disease cases. In contrast, Logistic Regression has an AUC of 94.6%. A curve closer to the top-left corner reflects better performance, and Random Forest clearly outperforms Logistic Regression in this area.

5. Balanced Accuracy:

- The balanced accuracy for Random Forest (0.9709) is higher than for Logistic Regression (0.8530), suggesting that while both models are effective, Random Forest offers a better overall balance between sensitivity and specificity.

Confusion matrices - Left (Random Forest), Right (Logistic Regression)

Confusion Matrix and Statistics

Prediction	Reference	
	Disease	No_Disease
Disease	111	6
No_Disease	0	97

Accuracy : 0.972
 95% CI : (0.94, 0.9896)
 No Information Rate : 0.5187
 P-Value [Acc > NIR] : < 2e-16

Kappa : 0.9437

Mcnemar's Test P-Value : 0.04123

Sensitivity : 1.0000
 Specificity : 0.9417
 Pos Pred Value : 0.9487
 Neg Pred Value : 1.0000
 Prevalence : 0.5187
 Detection Rate : 0.5187
 Detection Prevalence : 0.5467
 Balanced Accuracy : 0.9709

'Positive' Class : Disease

Confusion Matrix and Statistics

Prediction	Reference	
	Disease	No_Disease
Disease	101	21
No_Disease	10	82

Accuracy : 0.8551
 95% CI : (0.8007, 0.8994)
 No Information Rate : 0.5187
 P-Value [Acc > NIR] : < 2e-16

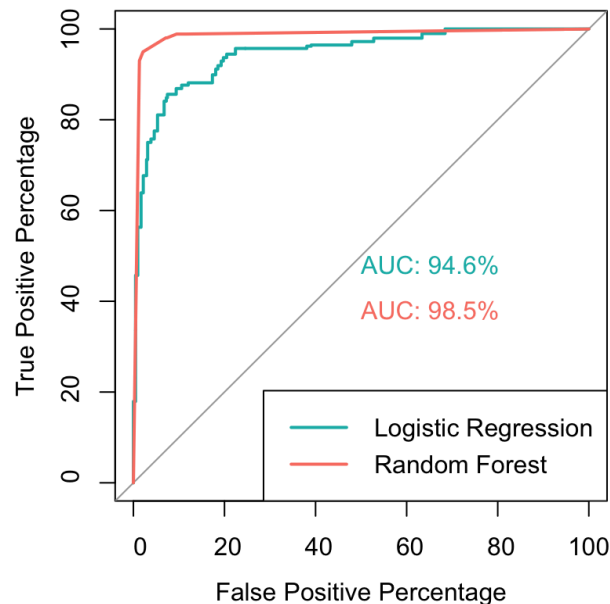
Kappa : 0.7088

Mcnemar's Test P-Value : 0.07249

Sensitivity : 0.9099
 Specificity : 0.7961
 Pos Pred Value : 0.8279
 Neg Pred Value : 0.8913
 Prevalence : 0.5187
 Detection Rate : 0.4720
 Detection Prevalence : 0.5701
 Balanced Accuracy : 0.8530

'Positive' Class : Disease

The Random Forest model outperformed the Logistic Regression model in both ROC and AUC metrics, demonstrating a superior ability to distinguish between classes. While the Random Forest model achieved better overall predictive accuracy, the Logistic Regression provided valuable interpretive insights into the effects of individual predictors, making it the preferred model for this problem.



Discussion and Conclusions

- This study compared Logistic Regression and Random Forest for predicting heart disease using a small dataset with 13 variables. Logistic Regression was chosen as the preferred model due to its high interpretability, providing quantifiable coefficients that clearly link individual predictors to the likelihood of heart disease. While Random Forest demonstrated slightly better sensitivity and specificity, Logistic Regression offers a straightforward approach to identify actionable factors, such as gender, chest pain type, maximum heart rate, major vessel coloring and thalassemia, etc. making it well-suited for real-world applications where understanding and explaining key risk factors is essential.
- The findings of this study align with Anshori, M., & Haris, M. S. (2022), where Logistic Regression was valued for its interpretability, computational efficiency, and practical use in clinical settings. Previous studies, such as Pal, M., & Parija, S. (2021) and Yazdani et al. (2021), have shown that Random Forest and more complex models like XGBoost often offer higher accuracy and better handling of complex data patterns, similar to the performance observed in this study. However, as highlighted in the literature, Logistic Regression remains a strong contender due to its simplicity and ability to provide meaningful insights into key risk factors, despite not achieving the same accuracy as more advanced models.

- This study is limited by its small dataset size, which may affect the generalizability of the results. Additionally, the model only predicts the presence or absence of heart disease, without addressing the severity of the disease, which is crucial for clinical decision-making
- Future research should refine models by exploring advanced techniques like XGBoost, Elastic Net Logistic Regression. Expanding data sources to include such as family history, lifestyle behaviors, and genetic predisposition will improve prediction accuracy. Additionally, predicting disease severity alongside diagnosis can enhance clinical applications by helping healthcare providers tailor treatment plans and make more informed decisions. Combining predictive power with interpretability will be key to developing actionable insights for healthcare.

References

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 - Pal, M., & Parija, S. (2021). Prediction of Heart Diseases using Random Forest. *Journal of Physics. Conference Series*, 1817(1), 12009-.
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-

Appendices

```
```{r}
library(tidyverse)
library(caret)
library(MASS)
library(kableExtra)
library(DataExplorer)
library(skimr)
library(psych)
library(forcats)
library(gridExtra)
library(outliers)
library(reactable)
library(ROSE)
library(mlogit)
library(ggplot2)
library(cowplot)
library(randomForest)
library(car)
library(pROC)
library(reshape2)
library(visdat)
library(psc1)
```
```

```
```{r}
set.seed(1234)
```
```

The data set is from Kaggle:

<https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>

```
```{r}
mydata<- read.csv("https://raw.githubusercontent.com/tonyCUNY/DATA_621/refs/heads/main/heart.csv")
```
```

```
```{r}
reactable(mydata)
```



```

...

EDA - Exploratory Data Analysis

Attribute Information:
age
sex
chest pain type (4 values)
resting blood pressure
serum cholesterol in mg/dl
fasting blood sugar > 120 mg/dl
resting electrocardiographic results (values 0,1,2)
maximum heart rate achieved
exercise induced angina
oldpeak = ST depression induced by exercise relative to rest
the slope of the peak exercise ST segment
number of major vessels (0-3) colored by flourosopy
thal: 0 = normal; 1 = fixed defect; 2 = reversible defect

```{r}
# Descriptive statistics
str(mydata)
...

```{r}
mydata %>%
 summary() %>%
 kable() %>%
 kable_styling()
...

```{r}
#Transform the variables as factor and numeric
mydata.clean <- mydata %>%
  mutate(
    sex = factor(ifelse(sex == 0, "F", "M")),
    age = as.numeric(age),
    trestbps = as.numeric(trestbps),
    chol = as.numeric(chol),

```

```

    thalach = as.numeric(thalach),
    cp = as.factor(cp),
    fbs = as.factor(fbs),
    restecg = as.factor(restecg),
    exang = factor(ifelse(exang == 0, "no", "yes")),
    slope = as.factor(slope),
    ca = as.factor(as.integer(ca)),
    thal = as.factor(as.integer(thal)),
    target = factor(ifelse(target == 0, "No_Disease", "Disease"))
  )
}

```{r}

Renaming columns
mydata.clean <- mydata.clean %>%
 rename(fasting_blood_sugar= fbs,
 exercise_induced_angina = exang,
 chest_pain = cp,
 resting_blood_pressure = trestbps,
 vessels_colored = ca,
 max_heart_rate = thalach)
}

```{r}

# Statistical summary
summary(mydata.clean)
}

```

Checking for Missing Value

1. No Missing Value for both numeric and Categorical Variables

```

```{r, echo=FALSE}
num_vars <- mydata.clean %>% select_if(where(is.numeric))
vis_miss(num_vars, cluster = TRUE) +
 ggtitle("Numeric Variables - No Missing Values") +
 theme(

```

```

 plot.title = element_text(face = "bold"),
 plot.margin = unit(c(1, 2, 1, 1), "cm")
)
 ...

  ```{r, echo=FALSE}
cat_vars <- mydata.clean %>% select_if(~ is.factor(.))
cat_vars <- cat_vars %>%
  mutate(across(everything(), ~na_if(as.character(.), "")))
vis_miss(cat_vars, cluster = TRUE) +
  ggtitle("Categorical Variables - No Missing Values") +
  theme(
    plot.title = element_text(face = "bold"),
    plot.margin = unit(c(1, 2, 1, 1), "cm")
  )
  ...

### Exploratory Plots of the Data

  ```{r, warning=FALSE}

cat_vars%>%
 gather() %>%
 ggplot(aes(value)) +
 geom_bar(fill = "lightblue", color="grey") +
 facet_wrap(~ key, scales = "free", ncol = 4) +
 theme(
 panel.grid = element_blank(),
 axis.text.x = element_text(angle = 45, hjust = 1)
) +
 labs(title = "Bar Plots of Categorical Variables")

 ...

  ```{r data_explorer, warning=FALSE, message=FALSE}
introduce(mydata.clean)

```

```

# par on plots
par(mfrow = c(1, 4))
plot_intro(mydata.clean)
describeBy(mydata.clean)
plot_histogram(mydata.clean)
plot_bar(mydata.clean)
...

```{r}
plot_bar(mydata.clean, by = "target")
...

```{r, echo=FALSE, message=FALSE, warning=FALSE}

# Boxplot
mydata %>%
  gather() %>%
  ggplot(aes(value)) +
  facet_wrap(~ key, scales = "free") +
  geom_boxplot(fill = "pink") +
  labs(title = "Box & Histogram Plots of Numeric Variables", x = "Value", y = "Frequency")

ggplot(gather(mydata), aes(value)) +
  facet_wrap(~ key, scales = "free") +
  geom_histogram(bins = 30, fill = "lightblue", color = "black")
...

```

The participants in the study are almost 70% male, and should be taken into consideration when used as an independent variable to predict heart disease:

```

```{r}
gender_data <- mydata.clean %>%
 group_by(sex) %>%
 summarise(sum_count = n()) %>%
 mutate(percentage = (sum_count / sum(sum_count)) * 100)

ggplot(gender_data, aes(x=factor(sex), y=sum_count, fill= factor(sex))) +
 geom_bar(stat = "identity")+

```

```

 geom_text(aes(label = paste0(round(percentage, 1), "%")),
 position = position_stack(vjust = 0.5), size = 5, color = "white") +
 labs(title="Percentage of Male and Female Study Participants", x="Gender", y="Frequency",
 fill="Gender")+
 scale_fill_manual(values= c("F" = "#CC79A7", "M"= "navyblue"), labels= c("Female", "Male"))+
 theme(panel.background = element_blank())
 #theme_bw()
 ...

  ```{r}
  ggplot(mydata.clean, aes(x=age, fill=factor(target))) +
  geom_histogram(bindwith = 4, position = "dodge", color = "black")+
    xlim(0,100)+
    ylim(0,80)+
  labs(title="Age Distribution of Participants With and Without Heart Disease", x="Age", y="Frequency",
       fill="Target Variable") +
  scale_fill_manual(values =c("No_Disease" = "blue", "Disease" = "red"), labels= c("Disease", "No
  Disease"))+
  facet_wrap(~target, scales = "free_y", labeller = labeller(target =
    c("No Disease" = "No Disease",
      "Disease" = "Disease")))+
  theme(panel.background = element_blank())
  ...

  ```{r}
 mydata.clean %>%

 group_by(chest_pain, target) %>%

 summarise(count = n()) %>%

 group_by(chest_pain) %>%

 mutate(percentage = (count / sum(count)) * 100) %>%

 ggplot(aes(x=chest_pain, y=count, fill= factor(target))) +
 geom_bar(stat = "identity")+
 geom_text(aes(label = paste0(round(percentage, 1), "%")),
 position = position_stack(vjust = 0.5), size = 5, color = "white") +
 labs(title="Frequency and Percentage of Participants With and Without Heart Disease \n at Each Chest

```

```

Pain Level", x="Chest Pain Levels", y="Frequency", fill="Target Variable")+
 scale_fill_manual(values= c("No_Disease" = "blue", "Disease"= "red"), labels= c("Disease", "No
Disease")))+

 # facet_wrap(~target, scales = "free_y", labeller = labeller(target =
c("0" = "No Disease",
"1" = "Disease")))+
 theme(panel.background = element_blank())

...

```{r}
mydata.clean %>%

  group_by(sex, target) %>%

  summarise(count = n()) %>%

  group_by(sex) %>%

  mutate(percentage = (count / sum(count)) * 100) %>%

  ggplot(aes(x=sex, y=count, fill= factor(target))) +
    geom_bar(stat = "identity")+
    geom_text(aes(label = paste0(round(percentage, 1), "%")),
              position = position_stack(vjust = 0.5), size = 5, color = "white") +
    labs(title="Frequency and Percentage of Participants With and Without Heart Disease for Males and
Females", x="Gender", y="Frequency", fill="Target Variable")+
    scale_fill_manual(values= c("No_Disease" = "blue", "Disease"= "red"), labels= c("Disease", "No
Disease")))+
    theme(panel.background = element_blank())

# mydata.clean %>%

#   # group_by(sex, target) %>%
#   #
#   # summarise(count = n())
#   ...

```

```

```{r}
mydata.clean %>%

 group_by(exercise_induced_angina, target) %>%

 summarise(count = n()) %>%

 group_by(exercise_induced_angina) %>%

 mutate(percentage = (count / sum(count)) * 100) %>%

 ggplot(aes(x=exercise_induced_angina, y=count, fill= factor(target))) +
 geom_bar(stat = "identity")+
 geom_text(aes(label = paste0(round(percentage, 1), "%"),
 position = position_stack(vjust = 0.5), size = 5, color = "white") +
 labs(title="Frequency and Percentage of Participants With and Without Heart Disease \n That Have and
Don't Have Exercised Induced Angina", x="Exercised-Induced Angina", y="Frequency", fill="Target
Variable")+
 scale_fill_manual(values= c("No_Disease" = "blue", "Disease"= "red"), labels= c("Disease", "No
Disease"))+
 theme(panel.background = element_blank())
```

```{r}
ggplot(mydata.clean, aes(x=resting_blood_pressure, fill=factor(target))) +
geom_histogram(bindwith = 4, position = "dodge", color = "black")+
 xlim(0, 230)+
 ylim(0,125)+
labs(title="Distribution of Resting Blood Pressure of Participants \n With and Without Heart Disease",
x="Blood Pressure", y="Frequency", fill="Target Variable") +
scale_fill_manual(values =c("No_Disease" = "blue", "Disease" = "red"), labels= c("Disease", "No
Disease"))+
facet_wrap(~target, scales = "free_y", labeller = labeller(target =
 c("No Disease" = "No Disease",
 "Disease" = "Disease")))+
 theme(panel.background = element_blank())
```

```{r}

```

```

ggplot(mydata.clean, aes(x=chol, fill=factor(target))) +
geom_histogram(bindwith = 4, position = "dodge", color = "black")+
 xlim(0,600)+
labs(title="Distribution of Cholesterol of Participants \n With and Without Heart Disease",
x="Cholesterol", y="Frequency", fill="Target Variable") +
scale_fill_manual(values =c("No_Disease" = "blue", "Disease" = "red"), labels= c("Disease", "No
Disease"))+
facet_wrap(~target, scales = "free_y", labeller = labeller(target =
 c("No Disease" = "No Disease",
 "Disease" = "Disease")))+
theme(panel.background = element_blank())
```



```

```{r}

ggplot(mydata.clean, aes(x=age, y=resting_blood_pressure, color=factor(target))) +
  geom_point()+
  xlim(0,90)+
  ylim(0,210)+
  #geom_bar()+
#geom_histogram(bindwith = 4, position = "dodge", color = "black")+
labs(title="Relationship Between Age and Resting Blood Pressure \n With and Without Heart Disease",
x="Age", y="Resting Blood Pressure", color="Target Variable") +
  scale_color_manual(values =c("No_Disease" = "blue", "Disease" = "red"), labels= c("Disease", "No
Disease"))+
#scale_fill_manual(values =c("0" = "blue", "1" = "red"), labels= c("No Disease", "Disease"))+
facet_wrap(~target, scales = "free_y", labeller = labeller(target =
  c("No Disease" = "No Disease",
    "Disease" = "Disease")))+
#scale_fill_manual(values =c("0" = "blue", "1" = "red"), labels= c("No Disease", "Disease"))+
theme(panel.background = element_blank())
```



```

```{r}

```


```


```



```

ggplot(mydata.clean, aes(x=chol, y=resting_blood_pressure, color=factor(target))) +
 geom_point()+
 xlim(0,500)+
 ylim(0,210)+
 #expand_limits(x=0,y=0)+
 #geom_bar()+
 #geom_histogram(bindwith = 4, position = "dodge", color = "black")+
 labs(title="Relationship Between Cholesterol and Resting Blood Pressure \n With and Without Heart
Disease", x="Cholesterol", y="Resting Blood Pressure", color="Target Variable") +
 scale_color_manual(values =c("No_Disease" = "blue", "Disease" = "red"), labels= c("Disease", "No
Disease"))+
 #scale_fill_manual(values =c("0" = "blue", "1" = "red"), labels= c("No Disease", "Disease"))+
 facet_wrap(~target, scales = "free_y", labeller = labeller(target =
 c("No Disease" = "No Disease",
 "Disease" = "Disease")))+
 #scale_fill_manual(values =c("0" = "blue", "1" = "red"), labels= c("No Disease", "Disease"))+
 theme(panel.background = element_blank())
...

```

## ## Correlation Matrix

A correlation matrix plot is a visual representation of the pairwise correlation coefficients between variables in your dataset. The values in the matrix represent the correlation coefficient (often Pearson's correlation) between two variables.

Interpret the Values:

High Positive Correlation (e.g., 0.8 or 0.9): Strong positive linear relationship between the two variables. For example, age and cholesterol levels might have a high positive correlation.

Moderate Correlation (e.g., 0.4 to 0.6): A moderate positive relationship. These variables are related, but not perfectly.

Low or No Correlation (e.g., 0.1 to 0.3 or close to 0): Weak or no linear relationship. These variables are not strongly associated.

Negative Correlation (e.g., -0.3 to -0.9): A negative relationship, where an increase in one variable corresponds to a decrease in the other.

```
``{r vif_model, warning=FALSE, message=FALSE}
```

```

cor_matrix <- cor(mydata.clean %>% select_if(where(is.numeric)), use = "complete.obs")

cor_long <- melt(cor_matrix)
cor_long <- cor_long[as.numeric(cor_long$Var1) > as.numeric(cor_long$Var2),]

ggplot(cor_long, aes(Var2, Var1, fill = value)) +
 geom_tile(color = "white") +
 geom_text(aes(label = ifelse(value != 0, round(value, 2), ""),
 color = "black", size = 3, face="bold")) + # Show only significant values
 scale_fill_gradient2(low = "pink", high = "blue", mid = "white",
 midpoint = 0, limit = c(-1, 1),
 space = "Lab", name = "Correlation") +
 theme_minimal() +
 theme(
 axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1, size = 10), # Adjust x-axis label
 axis.text.y = element_text(size = 10), # Adjust y-axis label
 axis.title = element_blank(), # Remove axis titles
 plot.title = element_text(hjust = 0.5, face = "bold", size = 14) # Center plot title
) +
 ggtitle("Correlation Matrix")
...

Data Balance

1. Target is fairly balanced (Almost 50-50% with no disease/disease)
2. Each level of categorical variables are represented by lots of patients
3. We can conclude the data is not imbalanced

```{r}

summary_data <- mydata.clean %>%
  group_by(target) %>%
  summarise(sum_count = n()) %>%
  mutate(percentage = (sum_count / sum(sum_count)) * 100)

ggplot(summary_data, aes(x = target, y = sum_count, fill = target)) +
  geom_bar(stat = "identity") +

```

```

    geom_text(aes(label = paste0(round(percentage, 1), "%")),
              position = position_stack(vjust = 0.5), size = 5, color = "white") +
    labs(title = "Sum Count of Target with Percentages",
         x = "Target Value",
         y = "Sum Count") +
    theme_minimal()
  ...

  ```{r}
 #
 variables <- c("sex", "chest_pain", "fasting_blood_sugar", "restecg", "slope", "vessels_colored")
 plot_data <- lapply(variables, function(var) {
 mydata.clean |>
 group_by(across(all_of(var)), target) |>
 summarise(count = n(), .groups = "drop") |>
 mutate(variable = var, level = get(var))
 }) %>%
 bind_rows()

 # Create the faceted bar plot
 ggplot(plot_data, aes(x = level, y = count, fill = factor(target))) +
 geom_bar(stat = "identity", position = "dodge") +
 facet_wrap(~variable, scales = "free_x") +
 labs(
 title = "Relationship between Target and Other Variables",
 x = "Variable Levels",
 y = "Count",
 fill = "Target"
) +
 theme_minimal() +
 theme(axis.text.x = element_text(angle = 45, hjust = 1))
 ...

 ## Splitting the data

  ```{r}
  # Partition data - train (80%) - test(20%)

```

```
set.seed(1234)
indexSet <- sample(2, nrow(mydata.clean), replace = T, prob = c(0.8, 0.2))
train <- mydata.clean[indexSet==1,]
test <- mydata.clean[indexSet==2,]
...
```

```
## Model 1 - Logistic Regression
```

```
```{r}
set.seed(1234)
logistic_full <- glm(target ~., data=train, family = "binomial")
summary(logistic_full)
...
```

```
VIF Check
```

VIF quantifies how much the variance of a regression coefficient is inflated due to multicollinearity with other predictors in the model. A high VIF indicates that a predictor is highly correlated with other predictors, meaning it might be redundant.

Multicollinearity in logistic regression can cause inflated standard errors for the coefficients, making them unreliable. This can lead to problems with statistical significance testing and interpretation.

High multicollinearity means that the model cannot determine the individual effect of correlated predictors, and you may end up with misleading results.

Interpretation of VIF Values:

VIF = 1: No correlation with other predictors (ideal).

$1 < \text{VIF} < 5$ : Moderate correlation; not a problem in most cases.

VIF > 5 or 10: High correlation with other predictors; indicates potential multicollinearity issues. It suggests that you should consider removing or combining predictors.

```
```{r}
set.seed(1234)
vif(logistic_full)
...
```

```
## Feature Selection - Improve the model with StepAIC
```

AIC (Akaike Information Criterion) or BIC (Bayesian Information Criterion) are generally better choices than Pseudo's R-squared.

AIC vs. BIC

AIC (Akaike Information Criterion):

Focuses on finding the model that best balances goodness-of-fit and complexity.

It penalizes the inclusion of extra predictors, but less strictly than BIC.

Use AIC if prediction accuracy is the primary goal, as it tends to favor slightly more complex models.

BIC (Bayesian Information Criterion):

Similar to AIC but applies a stronger penalty for model complexity, especially with larger sample sizes.

Use BIC if the focus is on selecting a parsimonious model (simpler with fewer predictors) or if your dataset is large.

We use AIC because our goal is prediction and we have more variables in this model. Step AIC function suggests us remove two variables: fbs, restecg.(which gives us the lowest AIC score after removing them)

```
```{r}
set.seed(1234)
logistic_2 <- stepAIC(logistic_full)
summary(logistic_2)
```

# Logistic Regression Coefficient (logit Model)
- The coefficient differ by a scale factor and therefore we cannot interpret the magnitude of the coefficients
- coefficient interpretation:

    Individual who are male, higher resting blood pressure, higher cholesterol, resting electrocardiographic results level 2, higher exercise induced angina, higher old peak, the slope of the peak exercise level 1, major vessels (0-3) colored by flourosopy level 1, 2, 3 are more likely to have heart disease.

- Odds Ratio higher than one would mean that the outcome of patients having heart disease,
```{r}
Odds Ratios
set.seed(1234)
exp(logistic_2$coefficients)
```

# Marginal Effect
```

The marginal effect refers to the change in the probability of the outcome (i.e., the dependent variable) associated with a one-unit change in a predictor variable, holding all other variables constant. Since logistic regression models the log-odds of the outcome, the marginal effect is a way to interpret the effect of a predictor variable in terms of actual probabilities, which is more intuitive.

```
```{r}
Logit model average marginal effects
set.seed(1234)

LogitScalar <- mean(dlogis(predict(logistic_2, type="link")))
avg_m_effects <- LogitScalar*coef(logistic_2)
avg_m_effects
```

# McFadden's Pseudo R-squared
```

Pseudo R-squared is a measure of model fit in logistic regression, similar to R-squared in linear regression, but it doesn't represent the proportion of variance explained. Instead, it quantifies how well the model explains the data compared to a null model (a model with no predictors).

McFadden's R-squared: Based on the log-likelihood ratio between the fitted model and the null model. Values closer to 1 indicate better fit.

The value implies that our model explains approximately 57.45% of the variation in the outcome variable compared to the null model. Values between 0.2 and 0.4 are often considered acceptable in practice, so a value of 0.59 suggests a very strong model fit.

Source:
["https://kapooramanpreet.github.io/documents/research/papers/Wolf,Kapoor,Hobson&Gardner-McCune_SIGCSE2023_Internships_Paper.pdf"](https://kapooramanpreet.github.io/documents/research/papers/Wolf,Kapoor,Hobson&Gardner-McCune_SIGCSE2023_Internships_Paper.pdf)
 McFadden, D. 1977. Quantitative Methods for Analyzing Travel Behaviour of Individuals: Some Recent Developments.

```
```{r}
set.seed(1234)

pseudo_r2 <- pR2(logistic_2)
pseudo_r2
```

# Predicting test data set and Accuracy.

1. We use K-fold cross validation
```

"In K-fold cross-validation, the data set is divided into a number of K-folds and used to assess the

model's ability as new data become available. K represents the number of groups into which the data sample is divided. For example, if you find the k value to be 5, you can call it 5-fold cross-validation."

2. Here we use 10 fold

3. Around 824 samples are used in training set and remaining 103 samples are used as test set.

4. Around 87% accuracy

```
```{r}
Assess model using Cross Validation
set.seed(1234)
crossValSettings <- trainControl(method = "repeatedcv", number = 10, savePredictions = TRUE)

crossVal <- train(target ~ age + sex + chest_pain + resting_blood_pressure + chol + max_heart_rate +
 exercise_induced_angina + oldpeak + slope + vessels_colored + thal, data=train, family = "binomial",
 method = "glm", trControl = crossValSettings)

crossVal

pred <- predict(crossVal, newdata = test)
confusionMatrix(data = pred, test$target)

```
```

Model 2 - Random Forest

1. A very low OOB (Out-of-bag) value suggest overfitting:

2. Reducing the complexity of the random forest model by decreasing the number of trees, limiting the maximum depth of trees, or removing features.

```
```{r}
set.seed(1234)
rf <- randomForest(target ~ ., data=train, proximity=TRUE, importance =TRUE, do.trace = 10, ntree = 50)
rf
```
```

1. Error rate become relatively flat after 5 trees

```
```{r}
Plot the error rate to see what is the suitable number of trees.
set.seed(1234)
```

```

oob.error.data <- data.frame(
 Trees=rep(1:nrow(rf$err.rate), times=3),
 Type=rep(c("OOB", "Disease", "No_Disease"), each=nrow(rf$err.rate)),
 Error=c(rf$err.rate[, "OOB"],
 rf$err.rate[, "Disease"],
 rf$err.rate[, "No_Disease"]))

ggplot(data=oob.error.data, aes(x=Trees, y=Error)) +
 geom_line(aes(color=Type))

```

```

...

```

1. Here we select 5 as number of tree
2. OOB is around 5% means 95% samples were correctly classified by the random forest

```

```{r}
set.seed(1234)
rf2 <- randomForest(target ~ ., data=train, proximity=TRUE, importance =TRUE, ntree = 5)
rf2
...

```

Feature Selection:

The function `varImpPlot()` in Random Forest regression (and classification) is used to visualize the importance of each predictor (independent variable) in the model. It helps to understand which features contribute most to the model's predictions. This is particularly useful when you have many predictor variables and want to identify which ones are most influential.

Purpose of `varImpPlot()` in Random Forest Regression:

Feature Importance: `varImpPlot()` plots the importance of each variable in the model, allowing you to assess which variables have the greatest effect on the target variable.

Model Interpretation: By visualizing feature importance, you can identify key predictors and gain insights into your data and model. This helps in understanding the model's behavior and making decisions about which features to retain, remove, or transform.

Dimensionality Reduction: If certain variables have very low importance, you might decide to drop them from the model to reduce complexity, improve interpretability, and potentially improve model performance.

Here we removed `fbs`, `restecg`, `slope` variables as they are less important.


```

```{r}
#Variable Importance - which variables can we remove?
set.seed(1234)
varImpPlot(rf2)
```

# Final model and accessing classification accuracy

```{r}
set.seed(1234)
rf3 <- randomForest(target ~ age + sex + chest_pain + resting_blood_pressure + chol + max_heart_rate +
 exercise_induced_angina + oldpeak + vessels_colored + thal, data=train, proximity=TRUE, importance
=TRUE, ntree = 5)
rf3

```

```{r}
set.seed(1234)
result <- data.frame(test$target, predict(rf3, test[, 0:13], type="response"))
head(result)
```

```{r}
set.seed(1234)
confusionMatrix(result$predict.rf3..test...0.13...type....response.., result$test.target)
```

```

Which model can better classify? -ROC Curve and AUC

To determine which model performs better we use AUC-ROC Curve:

AUC-ROC evaluates the ability of a model to distinguish between classes across various thresholds. A higher AUC indicates better discrimination ability. The model with the higher AUC is generally better.

Random Forest Model has higher AUC and below ROC Curve suggest Random Forest Model is better.

```
```\r}
set.seed(1234)
par(pty="s")
roc(train$target, logistic_2$fitted.values, plot=TRUE, legacy.axes=TRUE, percent=TRUE,
 xlab="False Positive Percentage", ylab="True Positive Percentage", col="#0ABAB5", print.auc=TRUE)
plot.roc(train$target, rf3$votes[,1], percent=TRUE, col="#FA8072", print.auc=TRUE, add=TRUE,
print.auc.y=40)
legend("bottomright", legend=c("Logistic Regression", "Random Forest"), col=c("#0ABAB5", "#FA8072"),
lwd=2)

...
```